Komunikační modul pro bezpilotní letoun

Adam Mikulič

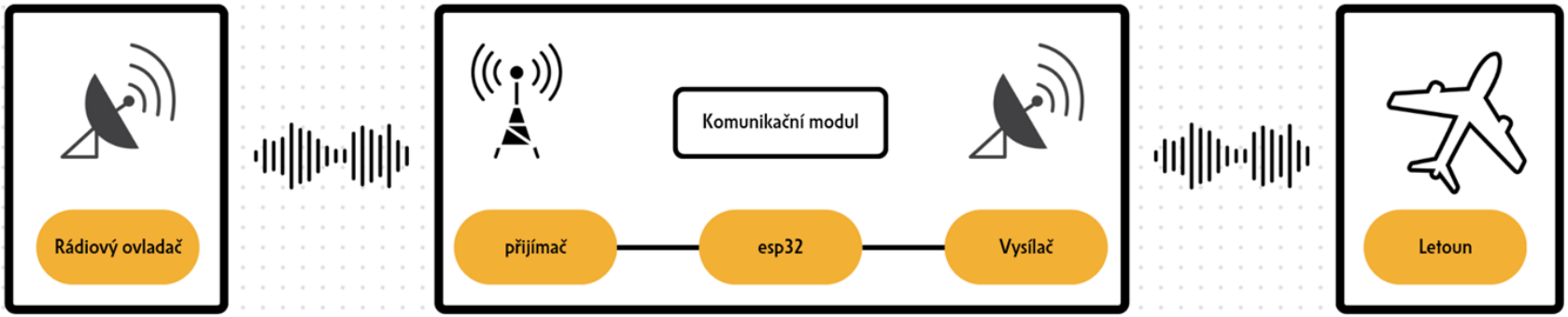
Definice problému

- omezení dosahu(EU limity výkonu, terén, ztráta přímé viditelnosti)
- vyšší výkon, směrové antény, komunikační modul

Cíl práce a kritéria

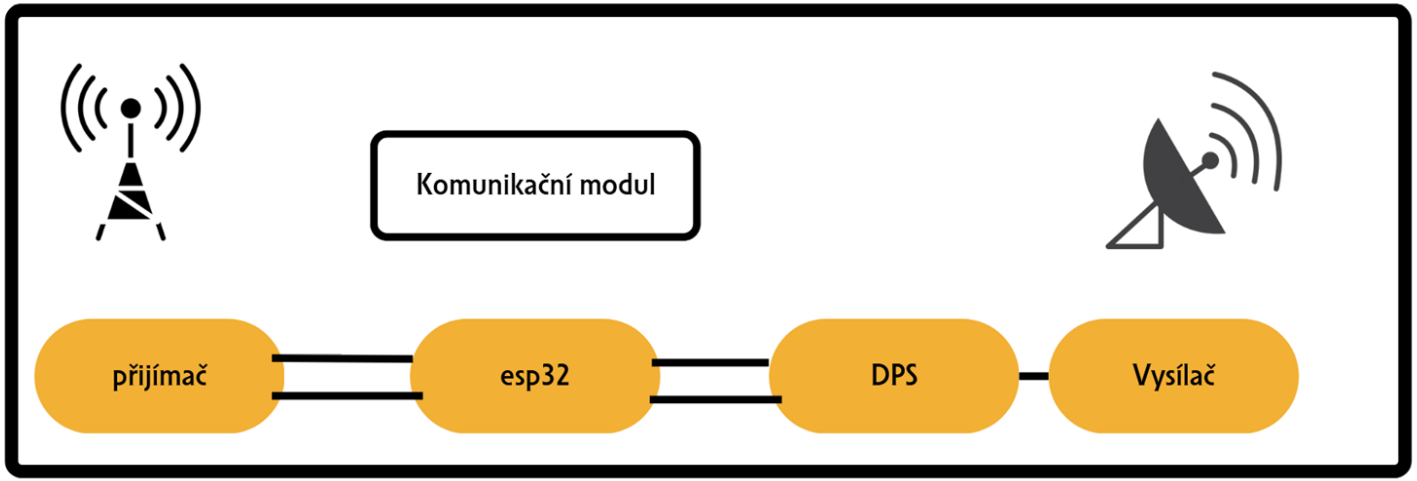
- návrh a sestavení komunikačního modulu
- stabilní komunikace
- minimální 2x rozšíření dosahu
- integrace do letounu + bezpečné vypuštění
- validace v reálných letových testech

Architektura systému



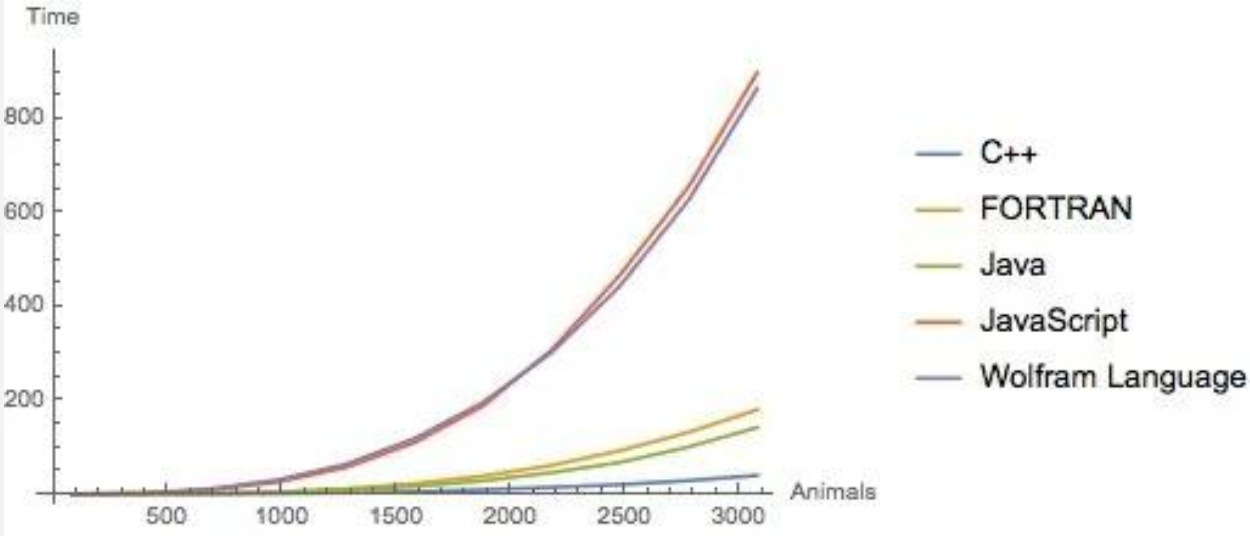
Architektura systému

- Half duplex x Full Duplex UART
- vlastní DPS jako řešení



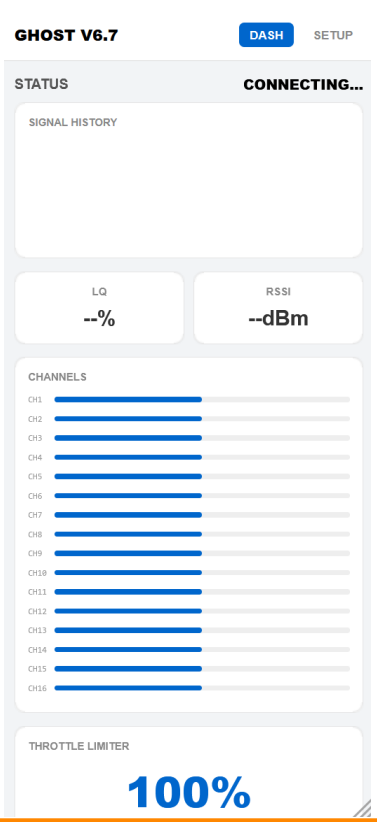
Softwarová integrace

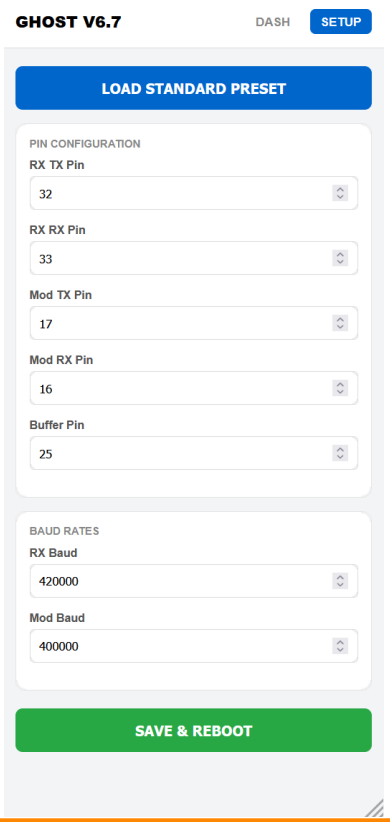
- C++
- FreeRTOS - multithreading (Dvě jádra - rádio a Wifi)



Softwarová integrace

- historie kvality signálu
- sledovač kanálů





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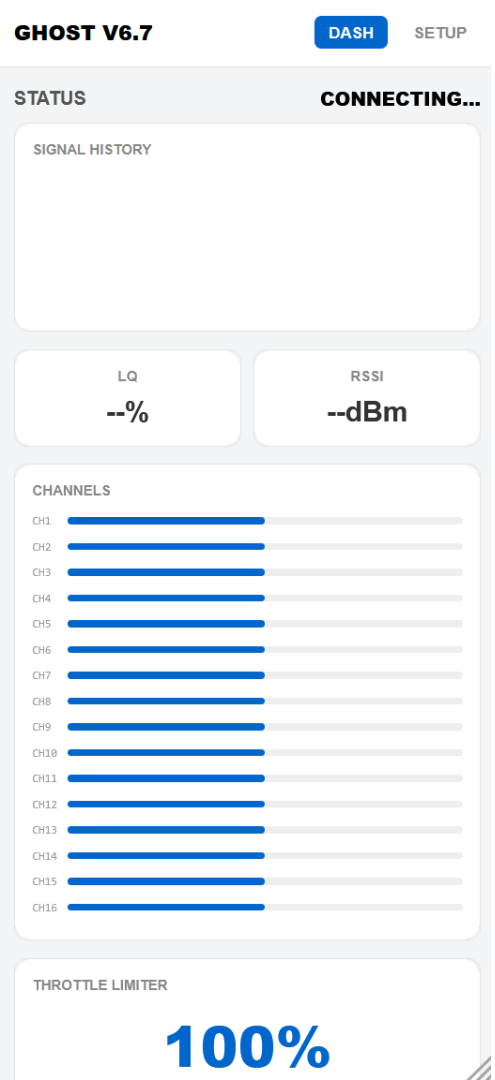
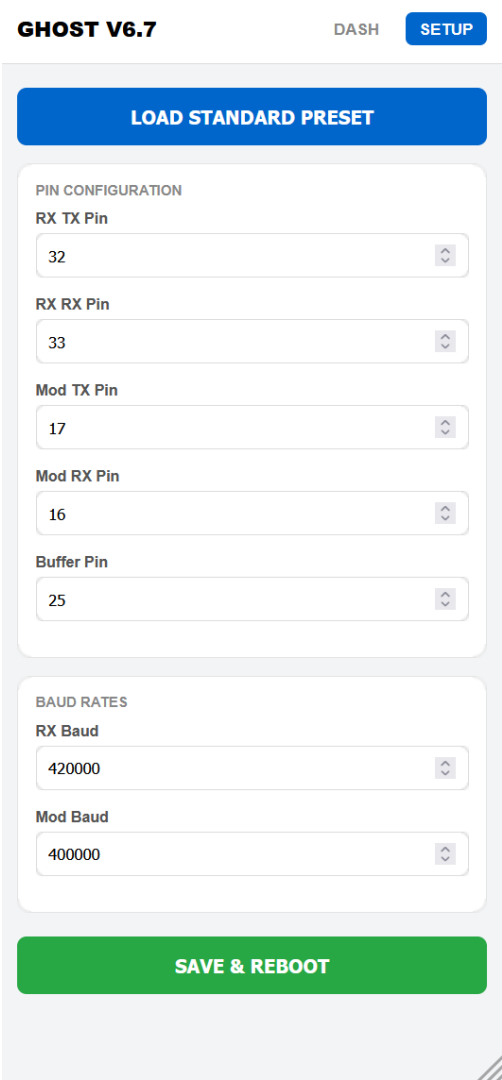
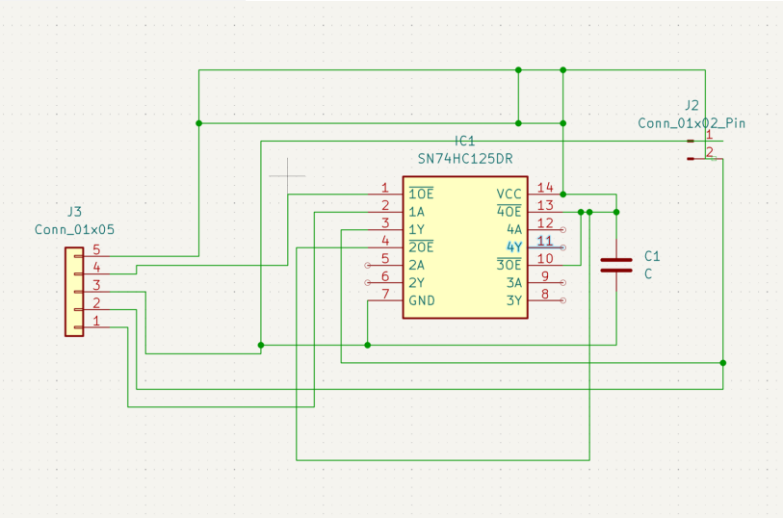
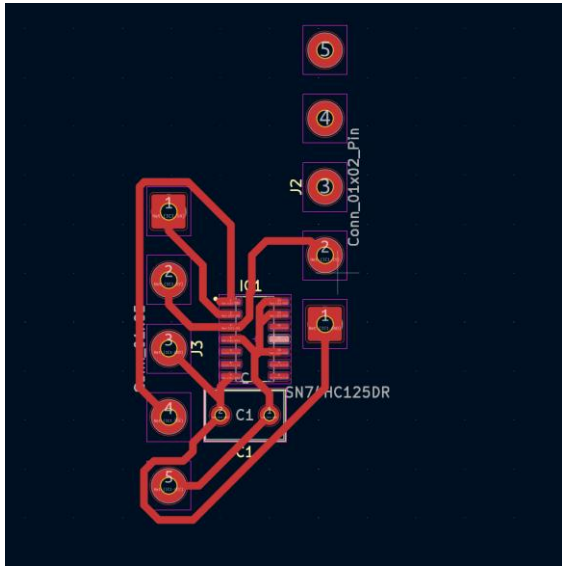




Mechanická integrace

- Vlastní DPS
- navrhnutý v programu KiCAD





Mechanická integrace

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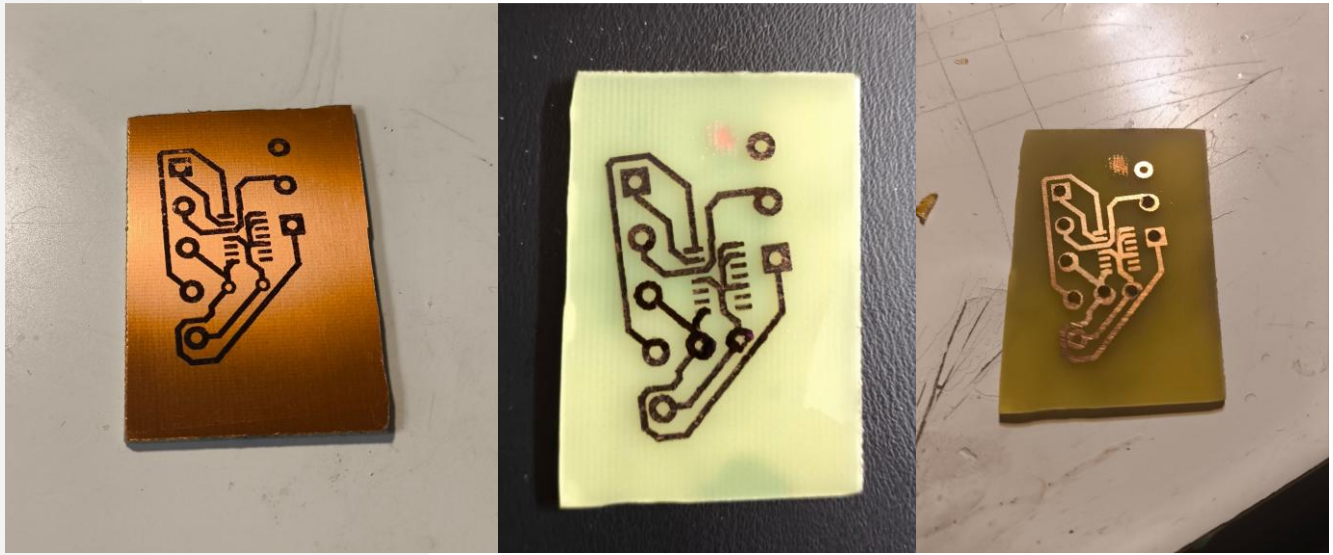

Mechanická integrace

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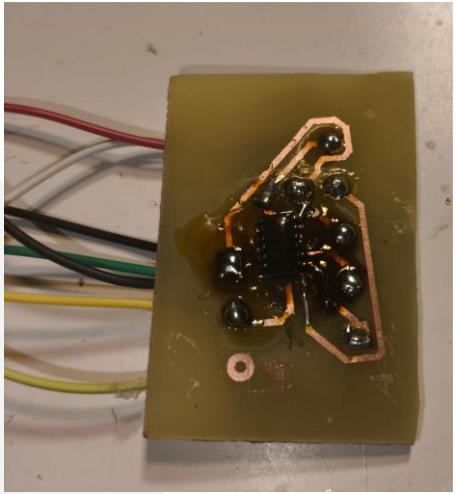
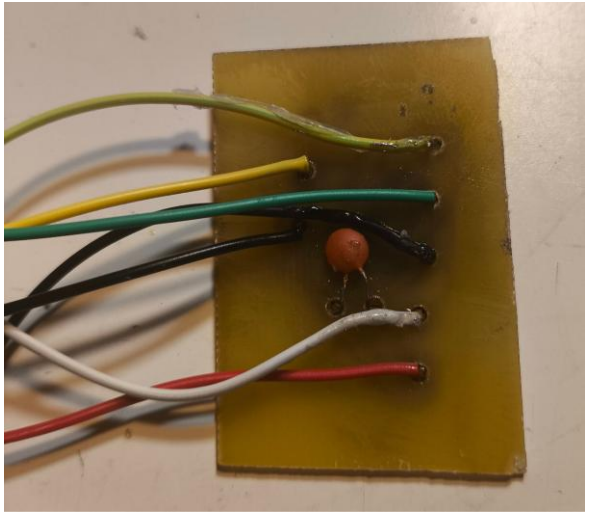
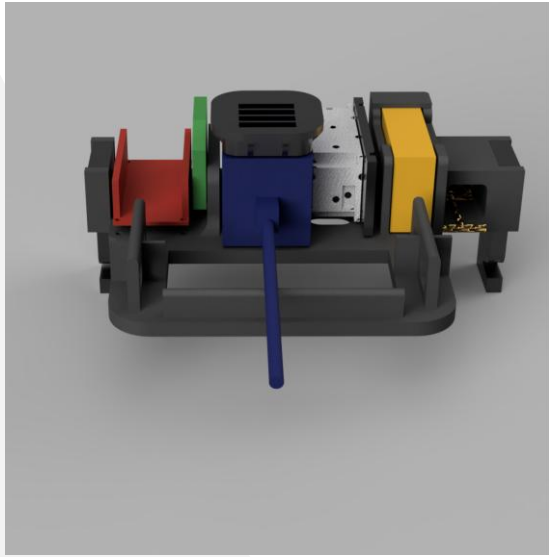
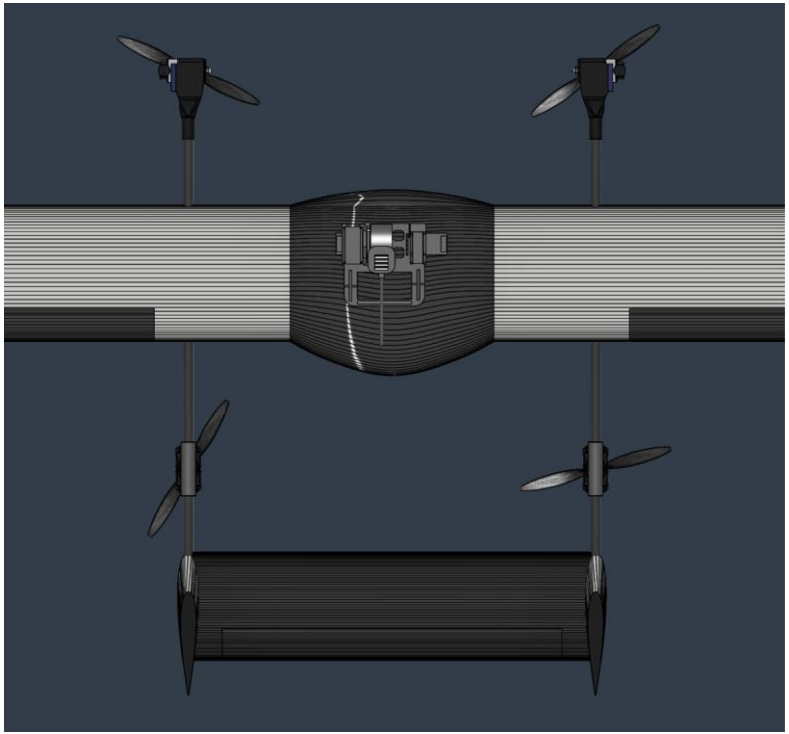




Mechanická integrace

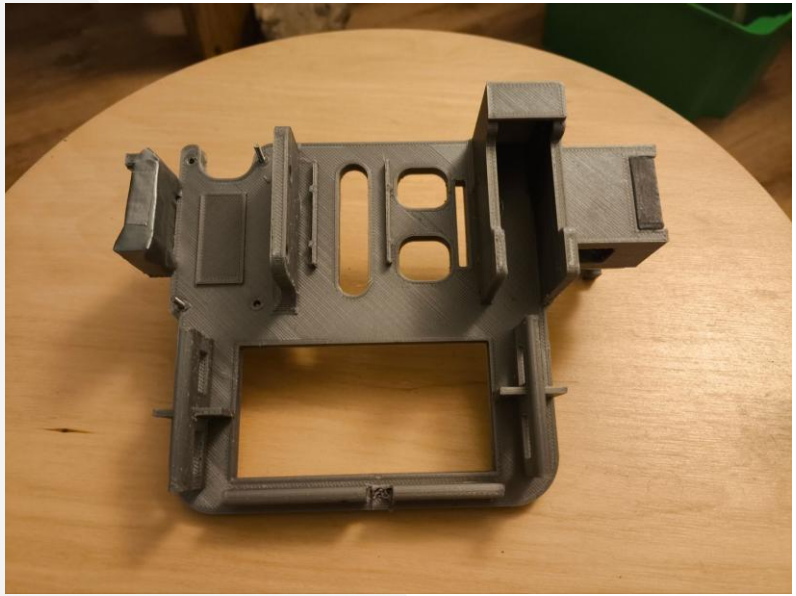
- Držák elektroniky
- navržen v Fusion 360

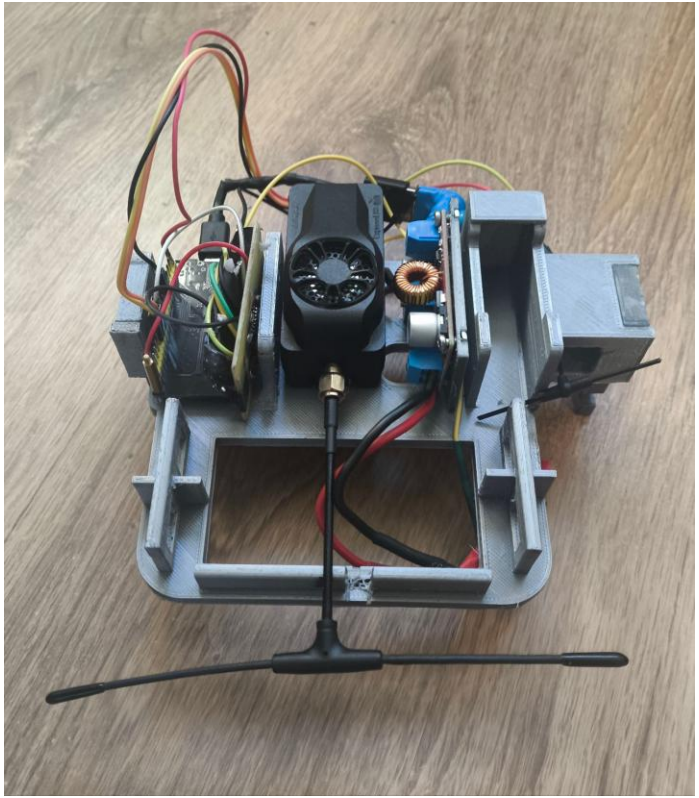




Mechanická integrace

- Držák elektroniky
- vytisknut na 3d tiskárně

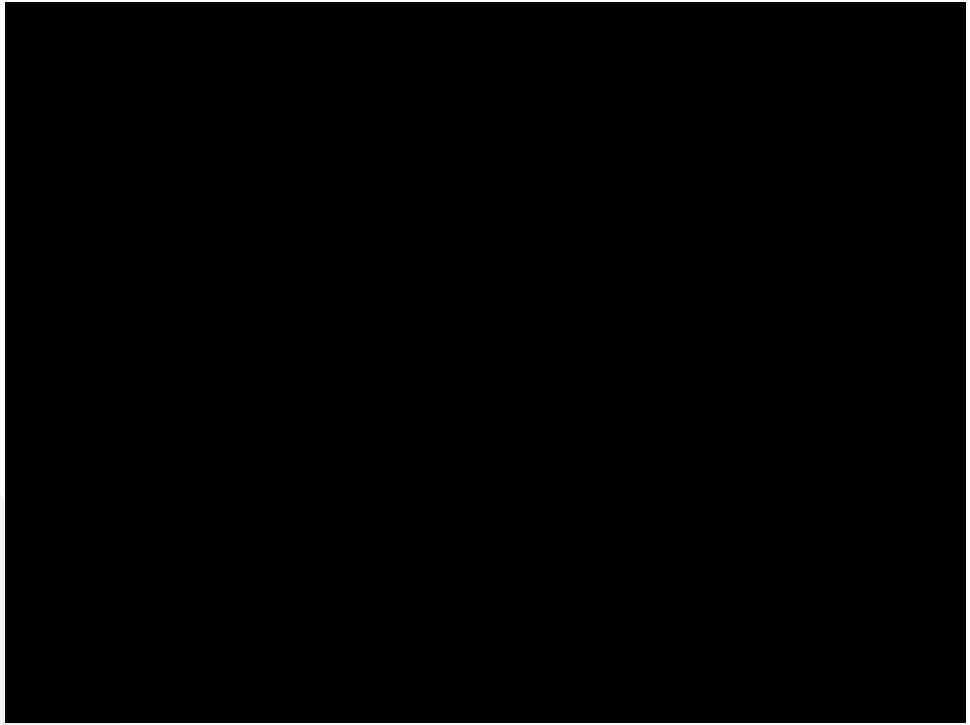
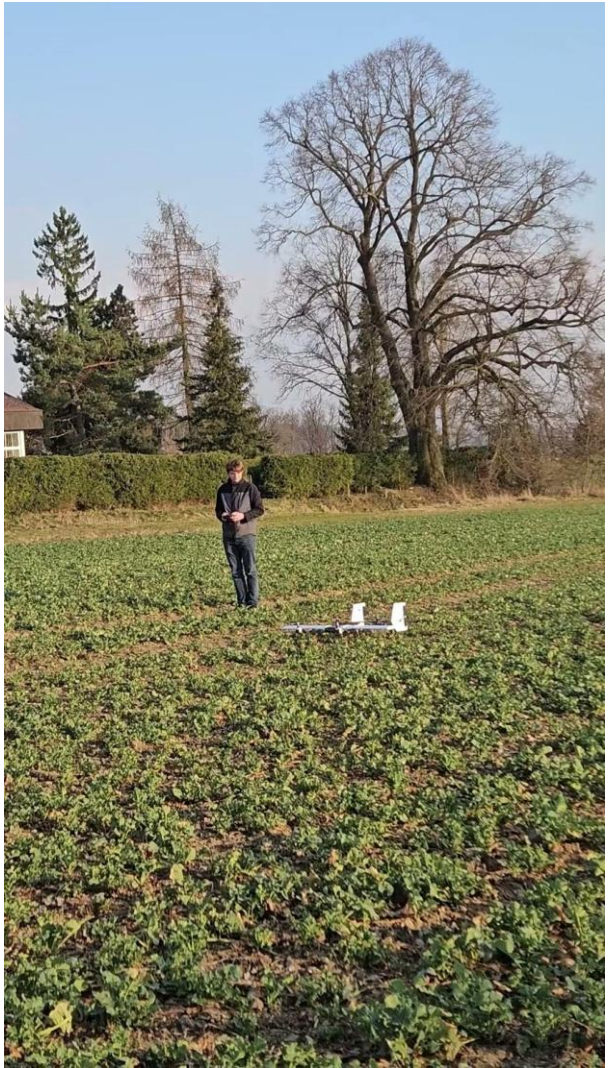




Mechanická integrace

- EPP obal
- navrhnut v Fusion 360
- vyříznut horkým drátem
- pokryt fólií

Testování



Děkuji za pozornost